

Program : Diploma in Cyber Forensics and Information Security	
Course Code : 6281A	Course Title: Virtualization and Cloud Security
Semester : 6	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L: 4 T: 0 P: 0)	Periods per semester: 60

Course Objectives:

- To learn and understand the benefits of virtualization.
- To give the information about the role of virtualization in cloud computing.
- To learn the knowledge about how security is provided in cloud computing.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic functions and features of Computer and Operating system.	3271	Operating Systems	3

Course Outcomes:

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Describe the concept of virtualization and its properties.	16	Understanding
CO2	Describe various types of virtualization	14	Understanding
CO3	Describe the architecture and characteristics of cloud computing	12	Understanding
CO4	Illustrate security mechanisms in cloud computing applications.	16	Understanding
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1		2	3				
CO2		2	3				
CO3	2	3	3				
CO4	2	3	3				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Describe the concept of virtualization and its properties.		
M1.01	Explain the concepts of virtualization and its architecture	3	Understanding.
M1.02	Illustrate the types of virtualization.	4	Understanding
M1.03	Describe the concept of virtual machines and its types	2	Understanding
M1.03	Explain the concept of Hypervisors.	2	Understanding
M1.04	Explain the types of hypervisors.	4	Understanding
M1.05	Describe the advantages and disadvantages of virtualization	1	Understanding
Contents: Overview of virtualization: - Concept of Virtualization, Virtualization Types - Desktop Virtualization - Network virtualization - Server and Machine Virtualization - Storage Virtualization - System-level or Operating Virtualization - Application Virtualization. Basics of virtual machine, Types of virtual machines. Hypervisor - Key Concepts-Types of Hypervisors. Virtualization Advantages and Disadvantages.			
CO2	Describe various types of virtualization		
M2.01	Illustrate the concept of Hardware virtualization	2	Understanding.

M2.02	Explain the concept of hardware assisted virtualization	2	Understanding
M2.03	Illustrate the benefits of hardware virtualization	1	Understanding
M2.04	Explain the concept of memory virtualization	2	Understanding.
M2.05	Describe the traditional address space	1	Understanding.
M2.06	Describe virtualized address space.	1	Understanding
M2.07	Explain virtualization techniques.	1	Understanding.
M2.08	Explain the concept of emulated TLB	2	Understanding.
M2.09	Explain shadow page tables	2	Understanding.
	Series Test - I	1	

Contents:

Hardware virtualization- Hardware assisted virtualization-Benefits of hardware based virtualization. Memory virtualization-Traditional address Space- Virtualized address space-virtualization techniques- Emulated TLB-Shadow page tables.

CO3	Describe the architecture and characteristics of cloud computing		
M3.01	Explain the concept of cloud computing and its architecture	2	Understanding
M3.02	Explain the characteristics of cloud computing.	1	Understanding
M3.03	Explain the advantage and the application of cloud computing	1	Understanding
M3.04	Describe the types of cloud computing	3	Understanding
M3.05	Explain the types of cloud services.	3	Understanding
M3.06	Explain the roles of virtualization in cloud computing	2	Understanding

Contents:

Cloud Computing Overview: -

Introduction-Architecture-Characteristics of Cloud Computing - Advantages of Cloud Computing-Applications, Cloud Computing Types –Public Cloud-Private Cloud-Hybrid Cloud, Cloud Services –Infrastructure as a Service (IaaS)-Platform as a Service (Paas)-Software as a Service (SaaS). Role of virtualization in cloud computing

CO4	Illustrate security mechanisms in cloud computing applications.		
M4.01	Explain the concept of cloud security architecture	2	Understanding
M4.02	Describe security Planning	1	Understanding
M4.03	Explain Cloud Security Alliance (CSA) model	2	Understanding
M4.04	Explain the concept of Data security and isolation of data	2	Understanding
M4.05	Explain infrastructure security services at cloud service providers	2	Understanding
M4.06	Explain application security services at cloud service providers.	2	Understanding
M4.07	Explain identity security services at cloud service providers	2	Understanding
M4.08	Explain identity security services in the enterprise	2	Understanding
M4.09	Describe the threats to a cloud security architecture	1	Understanding
	Series Test – II	1	
Contents: Cloud Security: - Security architecture – Security Planning, Cloud Security Alliances (CSA) model, Data Security-Secure isolation, Infrastructure Security Services – Application Security Services-Identity security services, Threats to a cloud security architecture			

Text / Reference

T/R	Book Title/Author
T1	Matthew Portnoy , Virtualization Essentials , 2 nd Edition
T2	A. Amali Angel Punitha, S. Sureshkumar, A. Peter Soosai Anandaraj, Introduction to Cloud Computing
T3	Ronald L. Krutz, Russell Dean Vines, Cloud Security: A Comprehensive Guide to Secure Cloud Computing
R1	Dan Kusnetzky, Virtualization: A Manager's Guide
R2	Chris Wolf , Erick M. Halter, Virtualization From the Desktop to the Enterprise
R3	Dac-Nhuong Le, Raghvendra Kumar, Nguyen Gia Nhu, Jyotir Moy Chatterjee, Cloud Computing and Virtualization.
R4	James Wilson, An Introduction to Cloud Computing
R5	Toby Velte, Anthony Velte, Robert Elsenpeter, Cloud Computing, A Practical

	Approach
R6	John W. Rittinghouse, James F. Ransome, Cloud Computing Implementation, Management, and Security

Online Resources

Sl.No	Website Link
1	https://www.vmware.com/in/solutions/virtualization.html
2	https://www.javatpoint.com/cloud-computing-security-architecture
3	https://www.tutorialspoint.com/cloud-security-architecture-and-types-of-cloud-computing-models
4	https://www.infoq.com/articles/cloud-security-architecture-intro/
5	https://www.guru99.com/cloud-computing-for-beginners.html